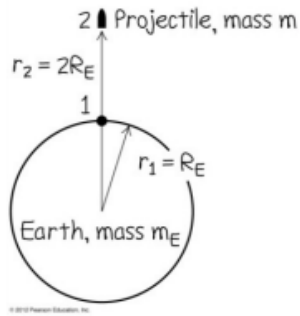
Example 13.5, P. 431 Escape speed



Shoot a projectile vertically with speed v_1, can it escape from earth's gravitational attraction?

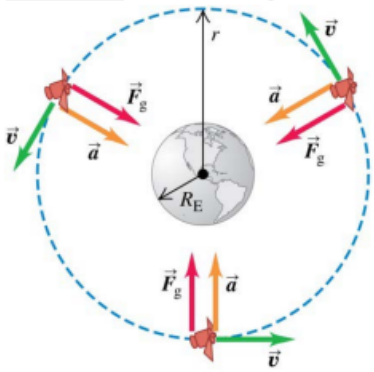
$$\tfrac{1}{2}mv_1^2 + \left(-\frac{Gm_E m}{R_E}\right) = 0 + 0 \longleftarrow \text{PE at } \infty$$

zero KE, i.e., correspond to minimum v_1

$$\Rightarrow \boxed{v_1 = \sqrt{\frac{2Gm_E}{R_E}}}$$

escape speed, ⚠ independent of m

Satellites – Assuming circular orbit



⚠ No tangential force, v must be constant

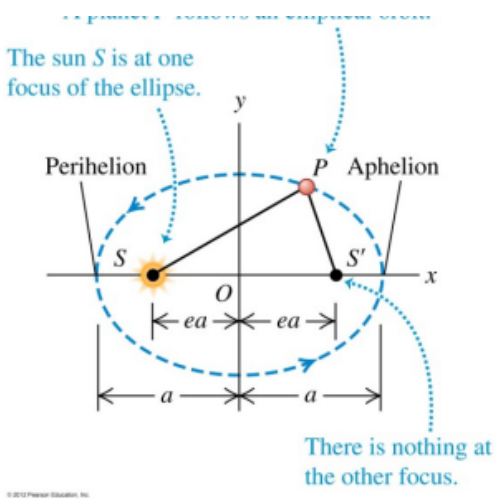
$$\frac{Gm_E m}{r^2} = m\underbrace{\left(\frac{v^2}{r}\right)}_{\text{centripetal acceleration}} \quad \Rightarrow \quad v = \sqrt{\frac{Gm_E}{r}}$$

⚠ do not confuse with escape speed

⚠ v independent of mass, astronauts orbit about the earth together with spacecraft – <u>apparent weightlessness</u>
<u>True weightlessness</u> only if object is infinitely far from other masses

Period of orbit $T = \dfrac{2\pi r}{v} = 2\pi r\sqrt{\dfrac{r}{Gm_E}} = \dfrac{2\pi}{\sqrt{Gm_E}}r^{3/2}$ ⚠ larger orbit ⟶ longer period

Total energy in an orbit $E = K + U = \tfrac{1}{2}mv^2 + \left(-\dfrac{Gm_E m}{r}\right) = \dfrac{1}{2}\left(-\dfrac{Gm_E m}{r}\right) = \dfrac{U}{2}$



An **ellipse** is defined by the locus of a point P such that $|PS'| + |SP| = \text{constant}$

S and S' are the two **foci** of the ellipse
Semi-major axis a (⚠ a length, not an axis)
Eccentricity e ($e = 0$ for circle, $0 < e < 1$ for ellipse)
Aphelion – farthest $[(1+e)a]$ point from sun
Perihelion – closest $[(1-e)a]$ point to sun
Note: aphelion distance + perihelion distance = $2a$

Second Law: A line from the sun to a given planet sweeps out equal areas in equal times.

Third Law: The periods of the planets are proportional to the $\frac{3}{2}$ powers of the major axis lengths of their orbits.

$$T = \frac{2\pi a^{3/2}}{\sqrt{Gm_S}}$$